

# Companion to Bird & de Moor, Algebra of Programming

## §8. Thinning Algorithms

### §8.1. Thinning

(8.1)

For  $Q : A \rightarrow A$ ,  $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$  (8.1).

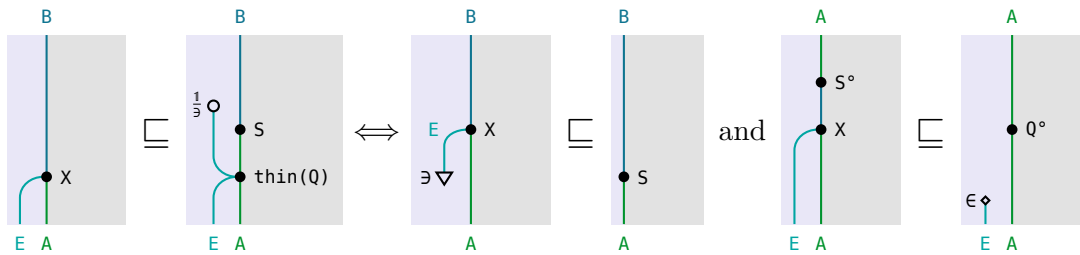
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$  the same  $^\circ$  as  $\triangleq \text{est}$ :  $\epsilon X$  runs  $a \rightarrow ys \rightarrow xs$ , member of  $ys$  first, and  $Q^\circ \epsilon$  runs  $a \rightarrow b \rightarrow xs$ , so  $(a, b) \in Q^\circ$  reads  $b \ Q \ a$ ; at  $Q^\circ \leq$  every  $a \in ys$  keeps some  $b \leq a$  in  $xs$ , the end  $\text{est}(\leq)$  picks

(8.1a)

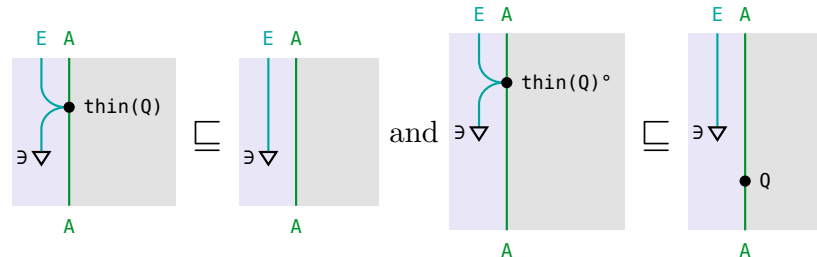
| the law                                                                                                                                                                                                                                                                                            | what it says                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \exists S$ and $S^\circ X \sqsubseteq Q^\circ \epsilon$                                                                                                                                                                             | everything kept is an S-value, and every S-value has a Q-lower bound among the kept ones                             |
| $\text{thin}(Q) \exists \sqsubseteq \exists$                                                                                                                                                                                                                                                       | everything a thinning keeps was in the set                                                                           |
| $\text{thin}(Q)^\circ \exists \sqsubseteq \exists Q$                                                                                                                                                                                                                                               | every element of the set has a Q-lower bound among the kept ones                                                     |
| $Q \sqsubseteq R \implies \text{thin}(Q) \sqsubseteq \text{thin}(R)$                                                                                                                                                                                                                               | the fewer pairs Q relates, the fewer subsets count as thinnings                                                      |
| $1 \sqsubseteq Q \implies 1 \sqsubseteq \text{thin}(Q)$<br>$Q \ Q \sqsubseteq Q \implies \text{thin}(Q) \ \text{thin}(Q) \sqsubseteq \text{thin}(Q)$                                                                                                                                               | keeping everything is always a legal thinning                                                                        |
| $Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge R \ R \sqsubseteq R \implies \text{est}(R) = \text{thin}(Q) \ \text{est}(R)$<br>$Q \sqsubseteq R, 1 \sqsubseteq Q, R R \sqsubseteq R$ — (8.1.2a); weaker than the book's "both preorders": Q transitive is never used and $1 \sqsubseteq R$ follows | <b>thin-introduction:</b> thinning first cannot lose an R-minimum                                                    |
| $\text{thin}(Q) \sqsupseteq \text{est}(Q) \ \frac{1}{\exists}$ (8.2) — (8.2)                                                                                                                                                                                                                       | <b>thin-elimination:</b> keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on |
| $\frac{S}{\exists} \text{thin}(Q) \sqsupseteq \frac{S}{\exists} \text{est}(R) \ \frac{1}{\exists}$<br>(8.3), $R \cap (S^\circ S) \sqsubseteq Q$ — thin-elimination variant                                                                                                                         | the usable variant: R need only refine Q between values S gives one argument                                         |
| $P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$ (8.4)                                                                                                                                                                                                                           | thinning each member set is a thinning of the union                                                                  |

#### §8.1.1. $X \sqsubseteq (S^\circ \exists) \text{thin}(Q) \iff X \exists \exists S$ and $S^\circ X \sqsubseteq Q^\circ \epsilon$

(8.1.1a)



(8.1.1b)



(8.1.1a) at  $X \triangleq \text{thin}(Q)$ ,  $S \triangleq \exists$ : the left side is  $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$ , since  $\exists \exists \exists = \Lambda(\exists) = 1$

#### §8.1.2. $\text{est}(R) = \text{thin}(Q) \ \text{est}(R)$ given $Q \sqsubseteq R$ , $1 \sqsubseteq Q$ , $R R \sqsubseteq R$

**$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$**  (8.1.2a)

the  $\sqsubseteq$  half: keeping everything is a thinning —  $1 \sqsubseteq Q$

(a) (b)

|     |                                 |                                         |
|-----|---------------------------------|-----------------------------------------|
| (a) | $\text{est}(R)$                 | $1 \sqsubseteq \text{thin}(Q)$ — (8.1a) |
| (b) | $\text{thin}(Q) \text{ est}(R)$ |                                         |

**$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$**  (8.1.2b)

the  $\sqsubseteq$  half, first condition of the UP of  $\text{est}$  at  $X \triangleq \text{thin}(Q) \text{ est}(R)$  —  $\text{est-up}$

(a) (b) (c)

|     |                                 |                                                         |
|-----|---------------------------------|---------------------------------------------------------|
| (a) | $\text{thin}(Q) \text{ est}(R)$ | $\text{est}(R) \sqsubseteq \exists$ — est-laws          |
| (b) | $\text{thin}(Q) \exists$        | $\text{thin}(Q) \exists \sqsubseteq \exists$ — (8.1.1a) |
| (c) | $\exists$                       |                                                         |

**$\epsilon \text{ thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$**  (8.1.2c)

the  $\sqsubseteq$  half, second condition —  $Q \sqsubseteq R$ ,  $R$  transitive

(a) (b) (c) (d) (e)

|     |                                           |                                                                    |
|-----|-------------------------------------------|--------------------------------------------------------------------|
| (a) | $\epsilon \text{ thin}(Q) \text{ est}(R)$ | $\epsilon \text{ thin}(Q) \sqsubseteq Q^\circ \epsilon$ — (8.1.1a) |
| (b) | $Q^\circ \epsilon \text{ est}(R)$         | $\epsilon \text{ est}(R) \sqsubseteq R^\circ$ — est-laws           |
| (c) | $Q^\circ R^\circ$                         | $Q \sqsubseteq R$                                                  |
| (d) | $R^\circ R^\circ$                         | $R$ transitive                                                     |
| (e) | $R^\circ$                                 |                                                                    |

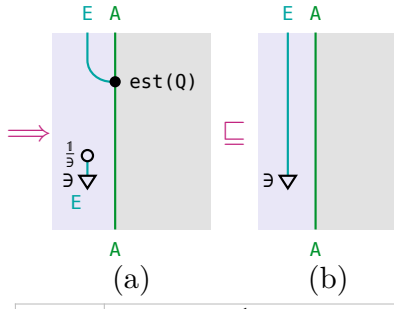
$$\text{est}(Q) \stackrel{1}{\exists} \sqsubseteq \text{thin}(Q)$$

(8.2)

the singleton holding a  $Q$ -least member of a set is a thinning of that set

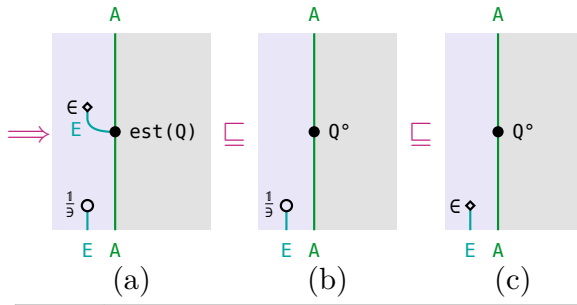
$$\text{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

every member of the singleton is a member of the set



|     |                                           |                                                                                        |
|-----|-------------------------------------------|----------------------------------------------------------------------------------------|
| (a) | $\text{est}(Q) \stackrel{1}{\exists} \ni$ | $\stackrel{1}{\exists} \ni = 1$ — pow-laws; $\text{est}(Q) \sqsubseteq \ni$ — est-laws |
| (b) | $\ni$                                     | $\triangleq \text{thin}, \epsilon \setminus (Q^\circ \epsilon)$ half                   |

$$\epsilon \text{ est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \epsilon$$

the singleton's member stands in  $Q$  to every member of the set

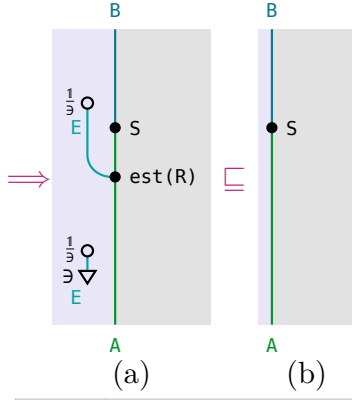
|     |                                                 |                                                         |
|-----|-------------------------------------------------|---------------------------------------------------------|
| (a) | $\epsilon \text{ est}(Q) \stackrel{1}{\exists}$ | $\epsilon \text{ est}(Q) \sqsubseteq Q^\circ$ — est-up  |
| (b) | $Q^\circ \stackrel{1}{\exists}$                 | $\stackrel{1}{\exists} \sqsubseteq \epsilon$ — pow-laws |
| (c) | $Q^\circ \epsilon$                              |                                                         |

$$R \cap S^\circ S \sqsubseteq Q \Rightarrow \frac{S}{S} \text{est}(R) \frac{1}{S} \sqsubseteq \frac{S}{S} \text{thin}(Q) \quad (8.3)$$

given  $R \cap (S^\circ S) \sqsubseteq Q$ ,  $Q$  a preorder

$$\frac{S}{S} \text{est}(R) \frac{1}{S} \ni \sqsubseteq S$$

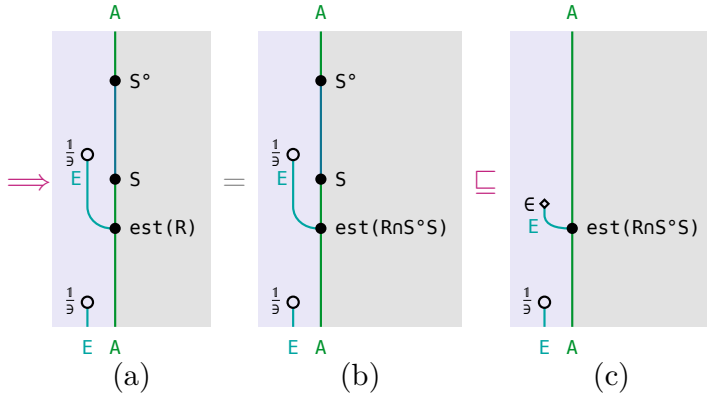
every member of the singleton is a value  $S$  returns



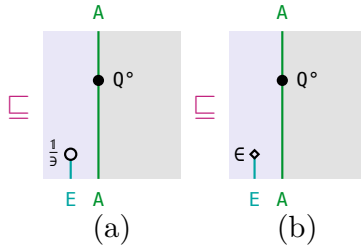
|     |                                             |                                                                                      |
|-----|---------------------------------------------|--------------------------------------------------------------------------------------|
| (a) | $\frac{S}{S} \text{est}(R) \frac{1}{S} \ni$ | $\frac{1}{S} \ni 1$ — pow-laws; $\frac{S}{S} \text{est}(R) \sqsubseteq S$ — est-laws |
| (b) | $S$                                         | (8.1a), second condition                                                             |

$$R \cap S^\circ S \sqsubseteq Q \Rightarrow S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S} \sqsubseteq Q^\circ \epsilon$$

the singleton's member stands in  $Q$  to every value  $S$  returns at the same argument



|     |                                                                |                                                                                                                           |
|-----|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| (a) | $S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S}$                | $\frac{S}{S} \text{est}(R) = \frac{S}{S} \text{est}(R \cap S^\circ S)$ — est-laws                                         |
| (b) | $S^\circ \frac{S}{S} \text{est}(R \cap S^\circ S) \frac{1}{S}$ | $S^\circ \frac{S}{S} \sqsubseteq \epsilon$ — pow-laws                                                                     |
| (c) | $\epsilon \text{est}(R \cap S^\circ S) \frac{1}{S}$            | $\epsilon \text{est}(R \cap S^\circ S) \sqsubseteq (R \cap S^\circ S)^\circ$ — est-up; $R \cap (S^\circ S) \sqsubseteq Q$ |



|     |                       |                                               |
|-----|-----------------------|-----------------------------------------------|
| (a) | $Q^\circ \frac{1}{S}$ | $\frac{1}{S} \sqsubseteq \epsilon$ — pow-laws |
| (b) | $Q^\circ \epsilon$    |                                               |

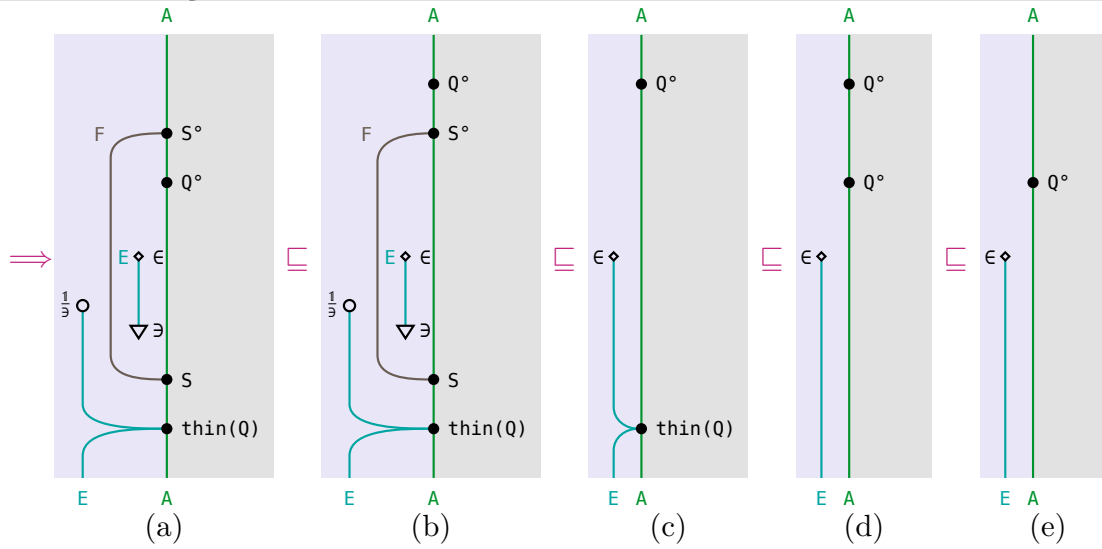
$$Q \subseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \subseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set —  $S$  monotonic on  $Q$ ,  $Q$  a preorder

$$\langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \exists \subseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \quad \text{and} \quad \langle \frac{(S)}{\exists} \rangle \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \subseteq Q^\circ \in$$

(8.1a) at  $X \triangleq \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle$ ,  $\langle \frac{(S)}{\exists} \rangle$  for its  $S$

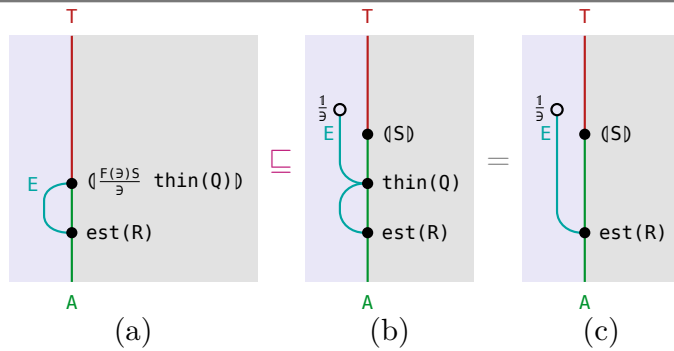


|     |                                                                      |                                                                         |
|-----|----------------------------------------------------------------------|-------------------------------------------------------------------------|
| (a) | $S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$ | $S^\circ F(Q^\circ) \subseteq Q^\circ S^\circ$ — mon-str, relator-laws  |
| (b) | $Q^\circ S^\circ F(\in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$ | $S^\circ F(\in) \frac{F(\exists) S}{\exists} \subseteq \in$ — pow-laws  |
| (c) | $Q^\circ \in \text{thin}(Q)$                                         | $\in \text{thin}(Q) \subseteq Q^\circ \in$ — $\triangleq$ thin, adj-all |
| (d) | $Q^\circ Q^\circ \in$                                                | $Q$ a preorder                                                          |
| (e) | $Q^\circ \in$                                                        |                                                                         |

$$Q \subseteq R \wedge 1 \subseteq Q \wedge Q \subseteq Q \wedge R \subseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \subseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{est}(R)$$

Corollary 8.1

the thinning fold refines the optimisation problem itself —  $S$  monotonic on  $Q$ ,  $Q \subseteq R$ , both preorders



|     |                                                                             |                                                         |
|-----|-----------------------------------------------------------------------------|---------------------------------------------------------|
| (a) | $\langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R)$ | thinning theorem                                        |
| (b) | $\langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{est}(R)$          | $\text{est}(R) = \text{thin}(Q) \text{est}(R)$ — (8.1a) |
| (c) | $\langle \frac{(S)}{\exists} \text{est}(R) \rangle$                         |                                                         |

## §8.2. Paths in a layered network

(8.2a)

$F(A, X) = A + A \times X$ ,  $L = \text{list}^+$  with initial algebra  $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$ .

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$ ,  $\text{consw}(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$ .

$\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$ ,  $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ .

$Q \triangleq R \cap (\text{head} \text{ head}^\circ)$ ,  $S \triangleq F(1, \exists) \alpha$ ,  $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$ ,  $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$ ,  $\text{step} \triangleq \text{cpr} P(\text{cons}) \text{est}(R)$ .

(8.2b)

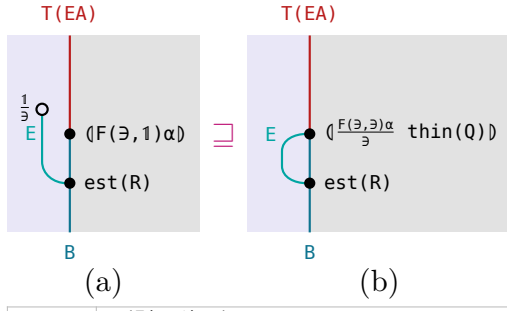
| the law                                                                       | what it says                                                                                                                                                           |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $F(\exists, Q) \alpha \subseteq F(\exists, 1) \alpha Q$                       | $F(\exists, 1) \alpha$ is monotonic on $Q$ ; on $R$ it is not, since the next edge can cost arbitrarily much                                                           |
| $S \text{ head} \subseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$ | $S \text{ head}$ is simple, which gives $R \cap (S^\circ S) \subseteq Q$ : between two paths $S$ builds from one argument, equal cost and equal head already means $Q$ |

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ \ R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(F(\exists, 1) \ \alpha, Q) \wedge R \sqcap (F(1, \exists) \ \alpha)^\circ \ F(1, \exists) \ \alpha \sqsubseteq Q \implies \quad (8.2c)$$

$$\langle \frac{F(\exists, 1)}{\exists} \ P(\frac{F(1, \exists)}{\exists} \ \alpha \ \text{est}(R)) \rangle \ \text{est}(R) \sqsubseteq \langle \frac{F(\exists, 1)}{\exists} \ \alpha \rangle \ \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers

$$= \frac{L(\exists)}{\exists} \ \text{est}(R)$$



|     |                                                                                                 |                       |
|-----|-------------------------------------------------------------------------------------------------|-----------------------|
| (a) | $\langle \frac{F(\exists, 1)}{\exists} \ \alpha \rangle \ \text{est}(R)$                        | Corollary 8.1, (8.2b) |
| (b) | $\langle \frac{F(\exists, \exists)}{\exists} \ \alpha \ \text{thin}(Q) \rangle \ \text{est}(R)$ |                       |

$\sqsubseteq$  (8.2d) under  $\langle \rangle$  monotonic: the whole chain runs inside the reduce, and the **est**(R) behind it never moves.

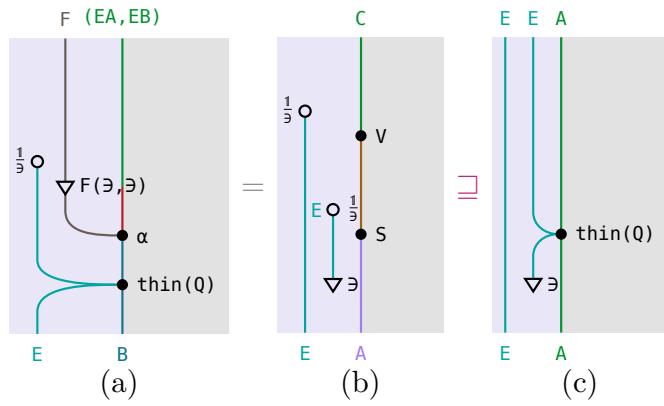
$$R \cap (F(1, \exists) \alpha)^\circ F(1, \exists) \alpha \subseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \subseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

$$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists}) \cup \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \subseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$$

thinning each set the algebra's parts build, then taking the union, is below thinning the one set the whole algebra builds

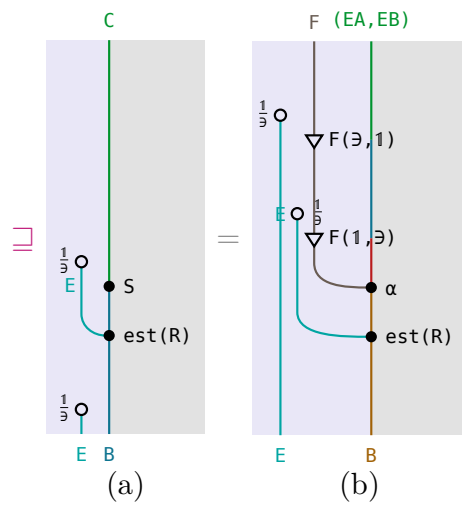
the algebra of (8.2c) row 2



|     |                                                             |                                                                                                                         |
|-----|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$ | $\frac{F(\exists, 1)F(1, \exists)\alpha = F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \cup$          |
| (b) | $\frac{V}{\exists} P(\frac{S}{\exists}) \cup$               | $\text{union thin}(Q) \supseteq P(\text{thin}(Q)) \cup \text{ — (8.1a)}$                                                |
| (c) | $P(\text{thin}(Q)) \cup$                                    | $\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \text{ — (8.1a), (8.2b)}$ |

$$R \cap (F(1, \exists) \alpha)^\circ F(1, \exists) \alpha \subseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \subseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists}) \cup P(\text{thin}(Q)) \cup$$

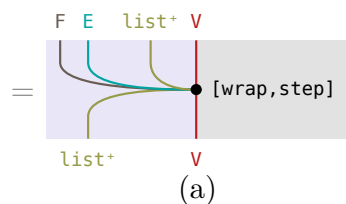
choosing an R-least member of each of those sets is below thinning each and taking the union



|     |                                                                                       |                                                                                                                                               |
|-----|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$                                   | $P(\frac{1}{\exists}) \cup \text{ — 1}$                                                                                                       |
| (b) | $\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R))$ | $\alpha \text{ a map; } \frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}] \text{ — } \triangle \text{ cost}$ |

$$\text{cp } P(\alpha) \text{est}(\text{path}) = [\text{wrap}, \text{step}]$$

at the layered network, choosing an R-least path from each case is the program's two cases  $[\text{wrap}, \text{step}]$



|     |                              |  |
|-----|------------------------------|--|
| (a) | $[\text{wrap}, \text{step}]$ |  |
|-----|------------------------------|--|

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \text{ — } \triangle \text{ cost}$$

## §8.3. Implementing thin

(8.3a)

$\text{setify} : [A] \rightarrow EA$ ,  $\text{cup} : EA \times EA \rightarrow EA$ ,  $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$ ,  $\text{listcp} : F(L) \rightarrow LF$ ,  $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P$  for  $P$  a connected preorder.

$\text{thinlist}(Q)$  is any  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$  with  $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$ ; one is  $\llbracket \text{nil}, \text{bump } Q \rrbracket$ ,  $\text{bump } Q (a, []) = [a]$ ,  $\text{bump } Q (a, [b] \# xs) = (b \sqsubseteq_Q a \rightarrow [a] \# xs, a \sqsubseteq_Q b \rightarrow [b] \# xs, [a] \# [b] \# xs)$ .

**Binary thinning** data:  $S = (f_1 p_1) \cup (f_2 p_2)$  with  $p_1, p_2$  coreflexive;  $Q$  a preorder with  $Q \sqsubseteq R$  and both  $f_1 p_1, f_2 p_2$  monotonic on  $Q$ ;  $P$  a connected preorder with both  $f_1, f_2$  monotonic on  $P$ ;  $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$ .

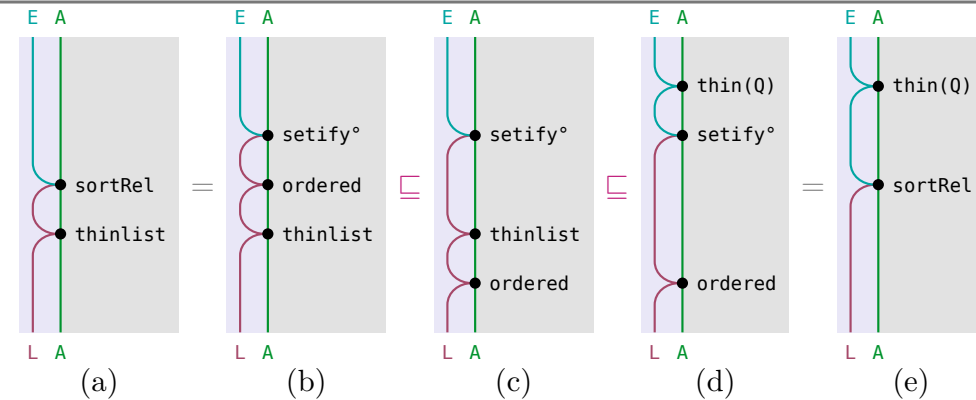
(8.3b)

| the law                                                                                                  | what it says                                                                                                       |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$<br>(8.5), $Q$ connected, $xs$ non-empty | what thinning should come to when it can: one element                                                              |
| $\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P)$ (8.6) — (8.6)            | thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys |
| $\text{sort}(P) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)                                      | a minimum of the sorted list is a minimum of the set                                                               |
| $\text{sort}(f P f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(P)$ (8.8)                        | shunt a function through a sort                                                                                    |
| $\text{sort}(P) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(P)$ (8.9), $p$ coreflexive               | filtering a sorted list sorts the restricted set                                                                   |
| $(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P)$ (8.10)          | merging two sorted lists sorts their union                                                                         |
| $F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$<br>(8.11), $F$ linear       | $\text{listcp}$ is the list implementation of the cartesian product $\text{cp}(F)$                                 |

$\text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered subseq} \sqsubseteq \text{subseq ordered} \wedge$  (8.6)

$\text{thinlist setify} \sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sortRel thinlist} \sqsubseteq \text{thin}(Q) \text{ sortRel}$

a thinning of the sorted list lists a thinning of the set —  $P$  a connected preorder,  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ .



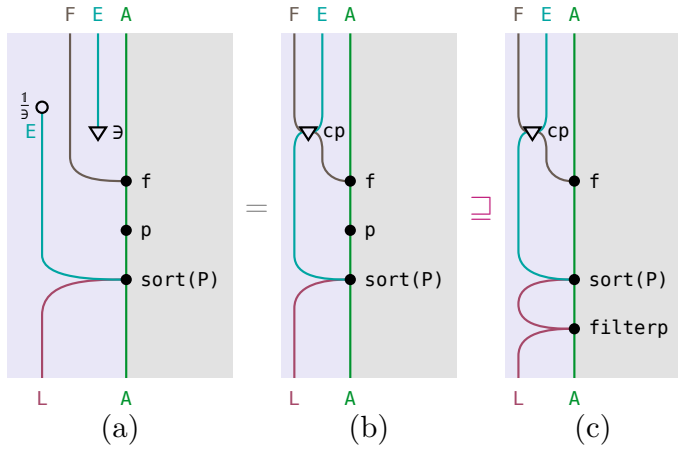
|     |                                                       |                                                                                                                                      |
|-----|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{sortRel thinlist}$                             | $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$                             |
| (b) | $\text{setify}^\circ \text{ ordered thinlist}$        | $\text{ordered } P \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ ordered } P \text{ — } \triangleq \text{thinlist}$      |
| (c) | $\text{setify}^\circ \text{ thinlist ordered}$        | $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q) \text{ — } \triangleq \text{thinlist, dom-laws, triple-chains}$ |
| (d) | $\text{thin}(Q) \text{ setify}^\circ \text{ ordered}$ | $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$                             |
| (e) | $\text{thin}(Q) \text{ sortRel}$                      |                                                                                                                                      |

**Map(f)  $\wedge$  ( $\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)$ )  $\wedge$  MonoAlg(f,P)  $\wedge$  sortF(f P f<sup>o</sup>) listf  $\sqsubseteq$  P(f) sort(P)  $\wedge$  sort(P) filterp  $\sqsubseteq$  E(p) sort(P)  $\wedge$  F(sort(P)) listcp  $\sqsubseteq$  cp sortF(F(P))  $\Rightarrow$  F(sort(P)) listcp listf filterp  $\sqsubseteq$   $\frac{F(\exists) f p}{\exists}$  sort(P)** Lemma 8.1

one sorted list built from sorted arguments, instead of a set built and then sorted — **f** : **FA**→**A** monotonic  
on **P**, **p** coreflexive, **F** linear.

Map(f)  $\wedge$  sort(P) filterp  $\sqsubseteq$  E(p) sort(P)  $\Rightarrow$  cp P(f) sort(P) filterp  $\sqsubseteq$   $\frac{F(\exists) f p}{\exists}$  sort(P)

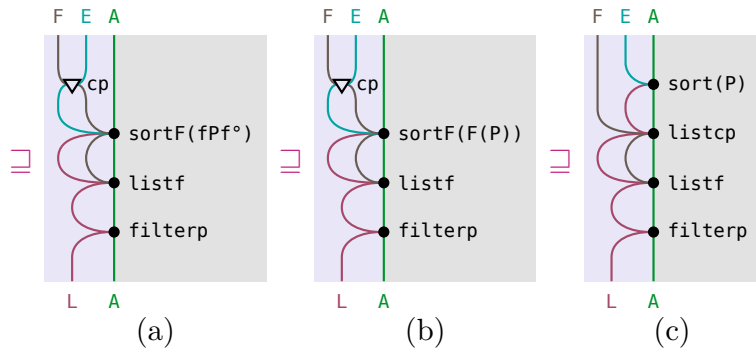
sorting the f-images of the product and filtering by p is below sorting the set F( $\exists$ )fp builds



|     |                                                  |                                                                                                                                              |
|-----|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{F(\exists) f p}{\exists} \text{ sort}(P)$ | $\frac{F(\exists) f p}{\exists} = \frac{F(\exists)}{\exists} E(fp)$ — pow-laws, $\triangleq$ thinlist<br>$E(f)=P(f)$ , f a map — powrel-laws |
| (b) | cp P(f) E(p) sort(P)                             | sort(P) filter(p) $\sqsubseteq$ E(p) sort(P) — (8.3b)                                                                                        |
| (c) | cp P(f) sort(P) filterp                          | sort(fPf <sup>o</sup> ) list(f) $\sqsubseteq$ P(f) sort(P) — (8.3b)                                                                          |

Map(f)  $\wedge$  ( $\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)$ )  $\wedge$  MonoAlg(f,P)  $\wedge$  sortF(f P f<sup>o</sup>) listf  $\sqsubseteq$  P(f) sort(P)  $\wedge$  F(sort(P)) listcp  $\sqsubseteq$  cp sortF(F(P))  $\Rightarrow$  F(sort(P)) listcp listf filterp  $\sqsubseteq$  cp P(f) sort(P) filterp

combining the arguments' sorted lists by listcp, mapping f and filtering is below sorting the f-images and filtering



|     |                                                  |                                                                                       |
|-----|--------------------------------------------------|---------------------------------------------------------------------------------------|
| (a) | cp sortF(f P f <sup>o</sup> ) listf filterp      | $FP \sqsubseteq fPf^o$ — mon-str, $\triangleq$ thinlist                               |
| (b) | cp sortF(F(P)) listf filterp                     | $F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$ — (8.3b) |
| (c) | $F(\text{sort}(P)) \text{ listcp listf filterp}$ |                                                                                       |

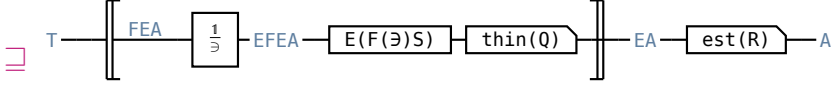
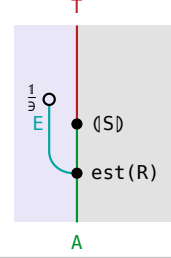
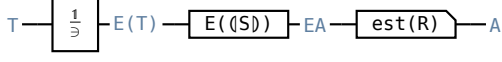
Theorem 8.2

$\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(f_1 \text{ } p_1, Q) \wedge$   
 $\text{MonoAlg}(f_2 \text{ } p_2, Q) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge$   
 $\text{MonoAlg}(f_2, P) \wedge \text{sortF}(f_1 \text{ } P \text{ } f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(P) \wedge \text{sortF}(f_2 \text{ } P \text{ } f_2^\circ) \text{ listf}_2 \sqsubseteq$   
 $P(f_2) \text{ sort}(P) \wedge \text{sort}(P) \text{ filterp}_1 \sqsubseteq E(p_1) \text{ sort}(P) \wedge \text{sort}(P) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(P)$   
 $\wedge F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp sortF}(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P) \wedge$   
 $\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P) \wedge \text{sort}(P) \text{ minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{listf}_1$   
 $\text{filterp}_1 \wedge g_2 = \text{listf}_2 \text{ filterp}_2 \wedge S = f_1 \text{ } p_1 \cup f_2 \text{ } p_2 \Rightarrow (\text{listcp } \langle g_1, g_2 \rangle \text{ merge}(P)$   
 $\text{thinlist}(Q)) \text{ minlist}(R) \sqsubseteq \frac{(S)}{\exists} \text{ est}(R)$

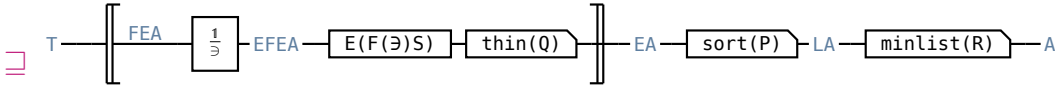
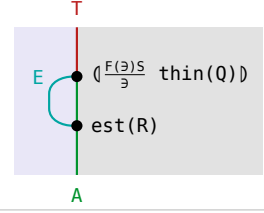
a fold on sorted lists of partial solutions, thinned at every step — at  $\triangleq \text{thinlist}$ 's binary thinning data.

circuit

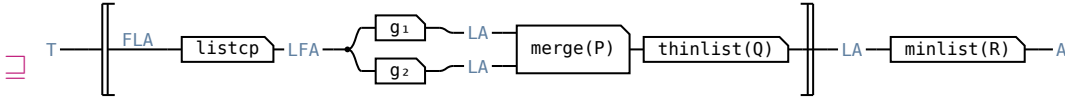
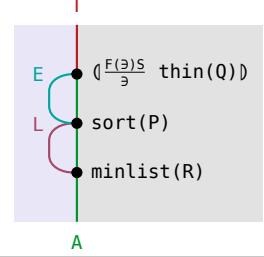
Hinze–Marsden



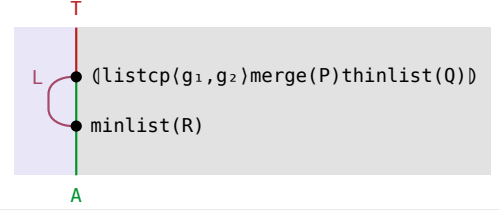
Corollary 8.1 at  $f_1 p_1$  and  $f_2 p_2$  monotonic on  $Q \triangleq \text{thinlist}$



$\text{sort}(P) \text{ minlist}(R) \sqsubseteq \text{est}(R) \text{ — (8.3b) at its } Q \triangleq R$



fusion at (8.3c)



$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \quad (8.3c) \\
& \text{sortF}(f_1 \ P \ f_1^\circ) \ \text{listf}_1 \sqsubseteq P(f_1) \ \text{sort}(P) \wedge \text{sortF}(f_2 \ P \ f_2^\circ) \ \text{listf}_2 \sqsubseteq P(f_2) \ \text{sort}(P) \wedge \text{sort}(P) \\
& \text{filterp}_1 \sqsubseteq E(p_1) \ \text{sort}(P) \wedge \text{sort}(P) \ \text{filterp}_2 \sqsubseteq E(p_2) \ \text{sort}(P) \wedge F(\text{sort}(P)) \ \text{listcp} \sqsubseteq \text{cp} \\
& \text{sortF}(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \ \text{merge}(P) \sqsubseteq \text{cup} \ \text{sort}(P) \wedge \text{sort}(P) \ \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \\
& \text{sort}(P) \Rightarrow F(\text{sort}(P)) \ \text{listcp} \ (\text{listf}_1 \ \text{filterp}_1, \text{listf}_2 \ \text{filterp}_2) \ \text{merge}(P) \ \text{thinlist}(Q) \sqsubseteq \\
& \quad \frac{F(\exists) \ (f_1 \ p_1 \cup f_2 \ p_2)}{\exists} \ \text{thin}(Q) \ \text{sort}(P)
\end{aligned}$$

sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                  | Hinze–Marsden |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |
| $\sqsubseteq$ <p><math>\text{sort}(P) \ \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \ \text{sort}(P) \text{ — (8.3b)}</math></p>                                                                                                                                                                                                                                                                                                       |               |
| $=$ <p><math>\langle \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \rangle \text{ cup } \text{sort}(P) \ \text{thinlist}(Q)</math><br/> <math>S = (f_1 p_1) \cup (f_2 p_2) \text{ — } \triangleq \text{thinlist, then } \triangleq \text{cup}</math></p>                                                                                                                                                          |               |
| $\sqsubseteq$ <p><math>\langle \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \rangle (\text{sort}(P) \times \text{sort}(P)) \ \text{merge}(P) \ \text{thinlist}(Q)</math><br/> <math>(\text{sort}(P) \times \text{sort}(P)) \ \text{merge}(P) \sqsubseteq \text{cup } \text{sort}(P) \text{ — (8.3b)}</math></p>                                                                                                  |               |
| $=$ <p><math>\langle \frac{F(\exists)f_1 p_1}{\exists} \ \text{sort}(P), \frac{F(\exists)f_2 p_2}{\exists} \ \text{sort}(P) \rangle \ \text{merge}(P) \ \text{thinlist}(Q)</math><br/> <math>\langle X, Y \rangle (\text{sort}(P) \times \text{sort}(P)) = \langle X \ \text{sort}(P), Y \ \text{sort}(P) \rangle \text{ — bdm-prod-laws}</math></p>                                                                                     |               |
| $\sqsubseteq$ <p><math>F(\text{sort}(P)) \ \text{listcp} \ \langle g_1, g_2 \rangle \ \text{merge}(P) \ \text{thinlist}(Q)</math><br/>     Lemma 8.1 at <math>f_1, p_1</math> and at <math>f_2, p_2</math>, then <math>X \langle g_1, g_2 \rangle \sqsubseteq \langle Xg_1, Xg_2 \rangle \text{ — bdm-prod-laws; } g_i \triangleq \text{list}(f_i)</math><br/> <math>\text{filter}(p_i) \text{ — } \triangleq \text{thinlist}</math></p> |               |

## §8.4. The knapsack problem

(8.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}, \text{ value} \triangleq \text{list}(\text{vol}) \ \text{sum}, \text{ weight} \triangleq \text{list}(\text{wt}) \ \text{sum}.$

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2])^\circ$ , within  $w$  the coreflexive on  $xs$  with weight  $xs \leq w, \ 0 \leq w$ .

$R \triangleq \text{value} \geq \text{value}^\circ, \ Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ), \ P \triangleq R, .$

$\text{FA} = 1 + \text{Item} \times A, \quad \text{listcp} = \text{wrap} + \text{cpr}, \quad g_1 \triangleq \text{list}([\text{nil}, \text{cons}]) \quad \text{filter}(\text{within} \quad w) \quad = [\text{list}(\text{nil}), h_1],$   
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2].$

$h_1 \triangleq \text{list}(\text{cons}) \ \text{filter}(\text{within} \ w), \ h_2 \triangleq \text{list}(\pi_2).$

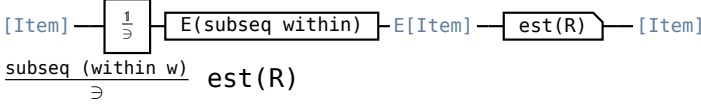
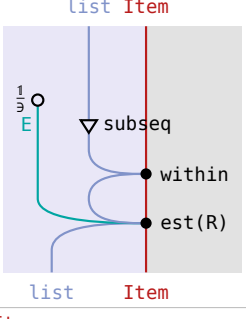
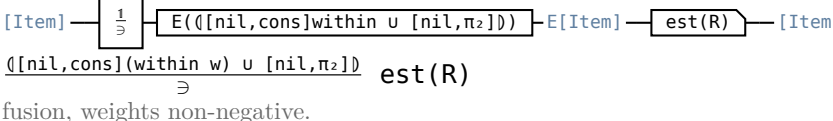
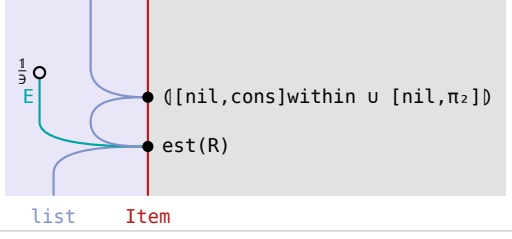
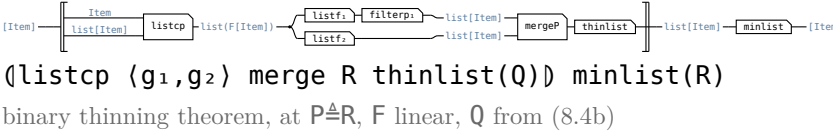
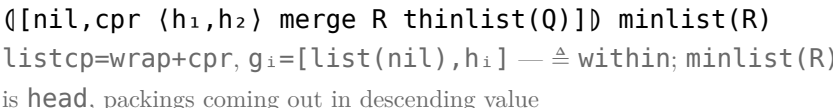
(8.4b)

| the law                                                                                                                                      | what it says                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| $(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$<br>FALSE                                      | not monotonic on R: a selection of greater value need not still fit once one more item goes in |
| $(1 \times Q) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) Q$<br>$(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$ , | both halves are monotonic on Q once ties in value are broken by weight                         |

(8.4c)

$$\begin{aligned}
& 0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{nil}, \text{cons}] R \\
& [\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sortP} \wedge \text{sortF}([\text{nil}, \pi_2] R [\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq P([\text{nil}, \pi_2]) \\
& \text{sortP} \wedge \text{sortP} \text{filterp}_1 \sqsubseteq E(\text{within}) \text{sortP} \wedge F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp sortF}(F(R)) \wedge (\text{sortP} \times \text{sortP}) \\
& \text{mergeP} \sqsubseteq \text{cup sortP} \wedge \text{sortP} \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt})) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(R) \implies \\
& ( \text{listcp } (\text{listf}_1 \text{filterp}_1, \text{listf}_2) \text{mergeP thinlist} ) \text{minlist} \sqsubseteq \frac{\text{subseq within}}{\supset} \text{est}(R)
\end{aligned}$$

the knapsack problem, as a fold that thins the packings kept at each item

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Hinze–Marsden                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  <p style="text-align: center;"><math>\frac{\text{subseq } (\text{within } w)}{\supset} \text{est}(R)</math></p>                                                                                                                                                                                                                                                                                                                |    |
|  <p style="text-align: center;"><math>\frac{([ \text{nil}, \text{cons} ] (\text{within } w) \cup [ \text{nil}, \pi_2 ])}{\supset} \text{est}(R)</math></p> <p style="text-align: center;">fusion, weights non-negative.</p>                                                                                                                                                                                                  |  |
|  <p style="text-align: center;"><math>( \text{listcp } (g_1, g_2) \text{merge } R \text{thinlist}(Q) ) \text{minlist}(R)</math></p> <p style="text-align: center;">binary thinning theorem, at <math>P^A R</math>, <math>F</math> linear, <math>Q</math> from (8.4b)</p>                                                                                                                                                     |  |
|  <p style="text-align: center;"><math>([ \text{nil}, \text{cpr } (h_1, h_2) \text{merge } R \text{thinlist}(Q) ] \text{minlist}(R))</math></p> <p style="text-align: center;"><math>\text{listcp} = \text{wrap} + \text{cpr}, g_i = [ \text{list}(\text{nil}), h_i ] \text{ --- } \triangleq \text{within}; \text{minlist}(R)</math></p> <p style="text-align: center;">is head, packings coming out in descending value</p> |                                                                                       |

## §8.5. The paragraph problem

(8.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[a]#xs, glue(a,xs)=[a]#head(xs)#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(8.5b)

| the law                                                                                                                                                                                                            | what it says                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| $(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$                                                                                                                                          | glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this |
| $(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$<br>$(1 \times Q) (\text{glue } (ok \ w)) \sqsubseteq \text{glue } (ok \ w) Q$<br>$\text{cons monotonic on } \text{collect} \sqsubseteq \text{collect}^\circ,$ | both halves are monotonic on Q once ties in waste are broken by the first line                                           |
| merge $T = \text{cat}$ ; $P \triangleq \text{head prefix head}^\circ$ also serves                                                                                                                                  | $T$ needs no sorting at all, and $\text{prefix}$ is a linear order on first lines of paragraphs of one input             |

$(\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap wrap,new}] \tau [\text{wrap wrap,new}]^\circ) \text{ listf}_1 \sqsubseteq P([\text{wrap wrap,new}]) \text{ sortP} \wedge \text{sortF}([\text{wrap wrap,glue}] \tau [\text{wrap wrap,glue}]^\circ) \text{ listf}_2 \sqsubseteq P([\text{wrap wrap,glue}]) \text{ sortP} \wedge \text{sortP filterp}_2 \sqsubseteq E(ok) \text{ sortP} \wedge F(\text{sortP}) \text{ listcp} \sqsubseteq \text{cp sortF}(F(T)) \wedge (\text{sortP} \times \text{sortP}) \text{ mergeP} \sqsubseteq \text{cup sortP} \wedge \text{sortP thinlist} \sqsubseteq \text{thin}(Q(\text{len},w)) \text{ sortP} \wedge \text{sortP minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp} (\text{listf}_1, \text{listf}_2 \text{ filterp}_2) \text{ mergeP thinlist}] \text{ minlist} \sqsubseteq \frac{\text{partition fits}}{\exists} \text{ est}(R))$

(8.5c)

a paragraph laid out as a fold that thins the layouts kept at each word

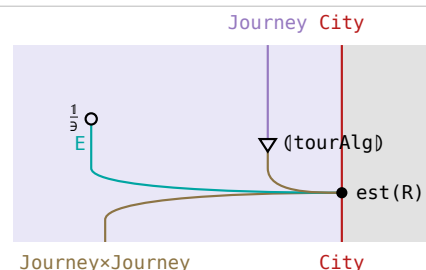
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Hinze–Marsden |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $\frac{\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} E(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap,new} u [\text{wrap wrap,glue}ok])} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))}{\text{partition list}^*(\text{fits } w)} \text{ est}(R)$                                                                                                                                               |               |
| $= \frac{\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} E(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap,new} u [\text{wrap wrap,glue}ok])} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))}{([\text{wrap wrap,new} u (\text{glue } (ok \ w))])} \text{ est}(R)$ <p>fusion, every word fits on a line by itself.</p>                                                                              |               |
| $= \frac{\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} E(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap,new} u [\text{wrap wrap,glue}ok])} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))}{([\text{wrap wrap,new} u ([\text{wrap wrap,glue} (ok \ w))])} \text{ est}(R)$ <p>the algebra as <math>(f_1 p_1) u (f_2 p_2)</math>, <math>p_1 \triangleq 1 \xrightarrow{\text{thinlist}}</math>.</p> |               |
| $\sqsubseteq ([\text{listcp} \langle g_1, g_2 \rangle \text{ cat thinlist}(Q)] \text{ minlist}(R))$ <p>binary thinning theorem, at <math>P \triangleq T</math> with merge <math>T = \text{cat}</math>, <math>Q</math> from (8.5b)</p>                                                                                                                                                                                                                                                   |               |
| $= [\text{start}, \text{cpr} \langle h_1, h_2 \rangle \text{ cat thinlist}(Q)]$ <p><math>\text{listcp} = \text{wrap} + \text{cpr}</math>, <math>g_i</math> along the coproduct <math>\xrightarrow{\text{partition}}</math></p>                                                                                                                                                                                                                                                          |               |

```
next $\hat{=}$ tail head, next2 $\hat{=}$ next $\times$ next, head2 $\hat{=}$ head $\times$ head, R $\hat{=}$ cost $\leq$ cost $^\circ$ , Q $\hat{=}$ Rn(next2 next2 $^\circ$ )n(head2 head2 $^\circ$ ), P $\hat{=}$ T,
g1 $\hat{=}$ list([start,dropl]), g2 $\hat{=}$ list([start,dropr]).
```

| the law                                                                                                                            | what it says                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| $(1 \times R) \text{ drop} \sqsubseteq \text{drop} \text{ } R$ FALSE                                                               | neither drop is monotonic on <b>R</b> : the two edges it adds and removes depend on <b>head</b> and <b>next</b> of both lists |
| $(1 \times Q) \text{ drop} \sqsubseteq \text{drop} \text{ } Q$<br>$(1 \times Q) \text{ drop} \sqsubseteq \text{drop} \text{ } Q$ , | both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input            |

a least-cost bitonic tour, as a fold that thins the tours kept at each city

Hinze–Marsden



The diagram shows a vertical line representing a boundary between two spaces: **Journey** (left, light blue background) and **City** (right, light grey background). A teal line labeled **list** originates from the Journey space and terminates at a point in the City space labeled  $(listcp(g_1, g_2)cat\ thinlist)$ . A blue line originates from the Journey space and terminates at a point in the City space labeled **minlist**. The bottom of the diagram is labeled **Journey x Journey** on the left and **City** on the right.

```

([start wrap,cpr (list(drop1),list(dropr)) cat
thinlist(Q)]) minlist(R)
listcp=wrap+cpr, gi=[list(start),list(dropi)] —  $\triangleq$  tour;
quadratic, two tours added per step

```